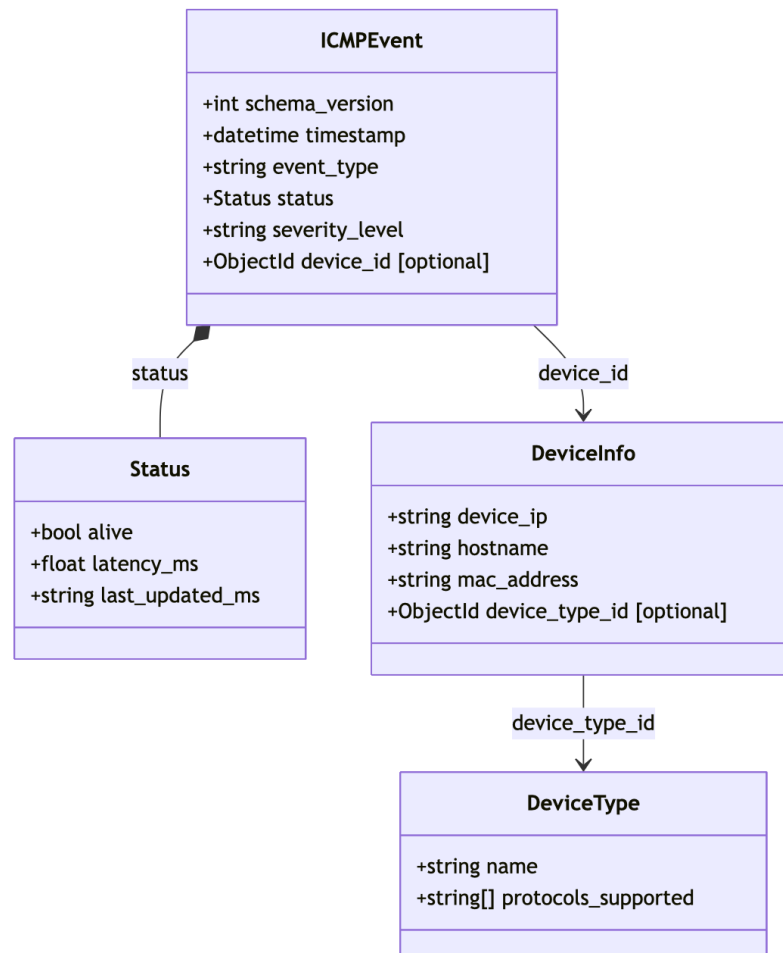


KIBBLE DATA MODEL

Overview

The Kibble monitors the network traffic and logs each check as an event. The data is kept in a normalized view with 3 collections; events, devices and device types. For any entry into the database, a unique id is created, which are used as the references between different collections.

Every device is associated with a device type, which determines what type of event we want to log for that device (for now ICMP is the only event protocol we are logging).



1. MongoDB- Kibble Database Organization

Collection	Purpose
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timeseries_events	Time-series of endpoint status events (up/down, latency, severity, etc.)
devices	Current snapshot of each monitored device (IP, hostname, etc.), keyed by IP.
device_types	Reference data: types of devices and which protocols they support.

2. timeseries_events Collection

Each monitoring cycle produces one event document per endpoint. The collection is a MongoDB time-series collection with:

-
- timeField: timestamp
- metaField: device_id
- granularity: seconds

Document Schema:

Field	Type	Description
schema_version	int	Version of the event schema (currently 1). Used for evolution and branching in consumers.
timestamp	datetime (UTC)	When the event occurred.
event_type	string	Either "endpoint_up" or "endpoint_down".
status	object	See below.
severity_level	string	One of: "debug", "info", "warning", "error", "critical".
device_id	ObjectId (optional)	Reference to devices._id when device normalization is used.

status Object:

Field	Type	Description
alive	bool	Whether the endpoint responded (e.g., ICMP reply received).

latency_ms	number	Round-trip latency in milliseconds (or timeout value if down).
last_updated_ms	number or string	When the reply was received (ms since epoch) or ISO timestamp.

Example Event:

```
{
  "schema_version": 1,
  "timestamp": "2025-02-15T12:00:00.000Z",
  "event_type": "endpoint_up",
  "status": {
    "alive": true,
    "latency_ms": 2,
    "last_updated_ms": "2025-02-15T12:00:00.002Z"
  },
  "severity_level": "info",
  "device_id": "507f1f77bcf86cd799439011"
}
```

3. devices Collection

This collection is refreshed at a configurable interval (for example, every 10 minutes).

Document Schema:

Field	Type	Description
device_ip	string	Endpoint IP address (primary key for upserts).
hostname	string	Hostname when available; otherwise default "".
mac_address	string	MAC address when available; otherwise default "".
device_type_id	ObjectId (optional)	Reference to device_types._id.

Example device:

```
{
  "device_ip": "10.128.0.1",
  "hostname": "server-a",
  "mac_address": "00:1a:2b:3c:4d:5e",
  "device_type_id": "507f1f77bcf86cd799439011"
}
```

4. device_type Collection

Reference table for categorizing devices and what protocols they support.

Document Schema:

Field	Type	Description
name	string	Human-readable type name (e.g. "device 1").
protocols_supported	array	List of protocol names (e.g. ["ICMP"]).

Example device type:

```
{
  "name": "device 1",
  "protocols_supported": ["ICMP"]
}
```